

Course Code	Course Title							
116U01C304	Object Oriented Programming Methodology							
	TH			P	TUT			Total
Teaching Scheme (Hrs./Week)	01			02	02*			05
Credits Assigned	01			01	02			04
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	30	20	50	25	--	--	25	150

* Batch wise Tutorial

Course prerequisites:

- Basics of Programming concepts

Course Objectives:

This course will provide the concept of object oriented designing and programming using JAVA and C++. These courses also provide differences in Object oriented programming approach in Java and C++. Students will learn about exception handling, Interfaces, file handling, Inheritance and Multithreading.

Course Outcomes:

At the end of successful completion of the course the student will be able to

- CO1. Understand the features of object oriented programming compared with procedural approach with C++ and Java
- CO2. Explore arrays, vectors, classes and objects in C++ and Java.
- CO3. Implement scenarios using object oriented concepts (Drawing class diagram, relationship between classes).
- CO4. Explore the interface, exceptions, multithreading, packages

Module No.	Unit No.	Details	Hrs.	CO
1	Fundamentals of Object oriented Programming		04	CO1
	1.1	Introduction, Procedural Programming Approach, Structured Programming Approach, Modular Programming Approach, OOP Approach		
	1.2	Objects and classes, Data abstraction and Encapsulation, Inheritance and Polymorphism, Runtime polymorphism, Static and Dynamic Binding, Exceptions, Reuse, Coupling and Cohesion, Object Oriented Features of Java and C++. Comparing Object Oriented Concepts with Java and C++		
2	Class, Object, Method and Constructor		08	CO1, CO2
	2.1	Class Object and Method: member, method, Modifier, Selector, iterator, State of an object, instanceof operator, Memory allocation of object using new operator.		
	2.2	Method overloading & overriding, constructor, destructor, Types of constructor (Default, Parameterized, copy constructor with object), Constructor overloading, this, final, super keyword, Garbage collection.		
3	Arrays String and vectors		09	CO2
	3.1	Arrays: Arrays: 1D, 2D, Variable Length array, for-each with Array, Array of objects, Vectors: Vector, ArrayList, Wrapper class. Command line Arguments.		
	3.2	Immutable string ,Methods of String class, String comparison, concatenation, substring, toString method		
	3.3	String-Buffer class, StringBuilder class		
4	Inheritance and Interface		08	CO1, CO4
	4.1	Inheritance Types of Inheritance, Final class, abstract class with constructor, abstract and non-abstract methods, super keyword, Method Overriding.		
	4.2	Interface, final keyword Implementing interfaces, extending interfaces Difference between an Abstract class and an Interface		

5	Class Diagram		06	CO1, CO 3
	5.1	Class Diagram		
	5.2	Implementing Aggregation and Association ,composition ,multiplicity ,Generalization		
6	Exception Handling & Packages, Multithreading		10	CO4
	6.1	Packages: Creating Packages, Using Packages, Access Protection, Predefined packages		
	6.2	Exception handling: Exception as objects, Exception hierarchy, Try catch finally Throw, throws		
	6.3	Multithreading: Thread life cycle, Multithreading advantages and issues, Simple thread program, Thread synchronization.		
Total			45	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Herbert schildt	<i>The complete Reference JAVA7</i>	Tata McGraw-Hill	7 th Edition 2017
2.	Kathy Sierra	<i>Sun Certified Programmer for JAVA</i>	McGraw-Hill Edition	6 th Edition, 2013
3.	Sachin Malhotra,Saurabh Chaudhary	<i>Programming in JAVA</i>	Oxford University	2 nd Edition, 2013
4.	E Balagurusamy	<i>Object Oriented Programming in C++</i>	Tata McGraw Hill	5 th Edition, 2011
5.	Grady Booch,James Rumbaugh,Ivar Jacobson	<i>Unified Modeling Language</i>	Pearson Education	3 rd Edition
6.	Yashwant Kanetkar	<i>Let us C++</i>	BPB publications	16 th Edition, 2020
7.	Ralph Bravaco,Shai simoson	<i>Java Programming from the Group up</i>	Tata McGraw-Hill	McGraw-Hill Edition